VAANYA KSHATRIYA

PUNE, MAHARSATRA

vaanya.kshatriya@gmail.com | +91 97111 02267 | linkedin.com/in/vaanya-kshatriya | github.com/vaanyas-codec

SUMMARY

Third-year B.Tech student in Computer Science (AI and Data Science) at MIT World Peace University, CGPA 9.22. I enjoy building ML pipelines, deploying full-stack applications, and leading student communities. Currently Vice President of Synapse AI Club and Thrive Head at E-Cell, with a research manuscript on fraud detection under review at the Journal of Young Investigators.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — **2024 to 2028**

Specialisation: Artificial Intelligence and Data Science | MIT World Peace University
Current CGPA: 9.22 | Semester GPA: 9.5

SKILLS

Languages	Python, C++, SQL, HTML, CSS, JavaScript
Frameworks and Tools	PyTorch, XGBoost, FastAPI, React, Node.js, Pandas, NumPy, OpenCV, Git, GitHub, Power BI
Core Competencies	Machine Learning, Computer Vision, Data Analysis, EDA, Data Visualisation, Full-Stack Development
Interests	AI/ML research, Building Data Products, Entrepreneurship and Leadership Roles

PROJECTS

Credit Card Fraud Detection — **2025 to Present**

Python, scikit-learn, XGBoost, Pandas, SMOTE | github.com/vaanyas-codec/credit-card-fraud-detection

- Built a leakage free ML pipeline on a severely imbalanced dataset (0.17% fraud rate) using SMOTE, feature scaling, and outlier capping. Benchmarked four classifiers; XGBoost achieved 91.23% F1 and 99.31% ROC-AUC, validated across 5-fold cross- validation and six metrics including PR-AUC.
- Built an interactive dashboard to visualise model performance and decision-threshold tradeoffs. Research manuscript under review at the Journal of Young Investigators.

Skin Lesion Classifier — **2026**

PyTorch, EfficientNet-B0, FastAPI, React, Grad-CAM | skin-lesion-classifier-seven.vercel.app

- Trained EfficientNet-B0 via transfer learning on HAM10000 (10K images, 58:1 class imbalance) for 7-class skin lesion classification. Found that accuracy-optimised training missed 40% of melanoma cases; switched to focal loss and improved melanoma recall from 60% to 72%.
- Used Grad-CAM saliency maps to confirm the model was attending to the lesion rather than background features. Deployed as a full-stack web app with a FastAPI backend and React frontend.

Real-Time ASL Detection **2026**

Python, OpenCV, MediaPipe, scikit-learn, Claude API | https://github.com/vaanyas-codec/realtime-sign-language-recognition

- Built a real-time ASL recognizer (webcam → MediaPipe landmarks → sklearn classifier) after a WLASL Transformer and a MediaPipe Model Maker attempt both fell short , the second blocked entirely by a broken dependency.
- Tracked down a live-accuracy collapse when the vocabulary scaled up by testing several hypotheses one by one, ruling out mirroring and feature-scale bugs before landing on the real cause: not enough variety in the training clips.
- Built a "collect mode" to record training data through the same live pipeline used for recognition, reaching 94.5% cross-validated accuracy across all 28 words.

EXPERIENCE

Intern at Accodyn Technologies **Feb 2025 to June 2025**

Frontend Development and Project Management

Worked across frontend development and project management for NBFC (Non-Banking Financial Company) clients, building responsive interfaces and coordinating with different teams to plan timelines, track deliverables, and translate business requirements into project plans.

EXTRACURRICULARS

Vice President, Synapse AI Club, MIT-WPU **2024 to Present**

Tech Team Member (2024), Treasurer (2025), Vice President (2026 onwards)

- Joined as a Technical Team Member in 2024, contributing to projects including DriveOpt. Became Treasurer in 2025, managing club finances across workshops, hackathons, and major events. Now serving as Vice President, leading club initiatives and technical sessions for a community of 500+ members.

Thrive Head, E-Cell, MIT-WPU **2024 to Present**

Member (2025), HOD Collaboration at MahaSummit, Core Organiser at RIDE, Thrive Head (2026 onwards)

- Joined E-Cell as a member in 2025, led Collaboration and Convention at MahaSummit, then joined the core organising body of RIDE as one of the main organisers. Managed end-to-end execution, mentor coordination, and participant milestones for the university's largest entrepreneurship event. This work contributed to MIT-WPU's recognition in the World Book of Records and the Asia Book of Records.

Member, Institution's Innovation Council (IIC) **2026 to Present**

Secretary, RIDE Division 1

- Contributing to campus innovation and entrepreneurship initiatives through the IIC; serving as Secretary for RIDE Division 1.